Homework 9

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

Contents

Problem 1 (*Ross Chapter 3 17.1*)

Let $f(x) = \sqrt{4-x}$ for $x \le 4$ and $g(x) = x^2$ for all $x \in \mathbb{R}$.

(a) Give the domains of $f+g$, fg, $f \circ g$, and $g \circ f$.

(b) Find the values $f \circ g(0)$, $g \circ f(0)$, $f \circ g(1)$, $g \circ f(1)$, $f \circ g(2)$, and $g \circ f(2)$.

(c) Are the functions $f \circ g$ and $g \circ f$ equal?

(d) Are $f \circ g(3)$ and $g \circ f(3)$ meaningful?

Discussion.

Solution. (a) We are given

$$f(x) = \sqrt{4-x} \quad (x \le 4), \qquad g(x) = x^2 \quad (x \in \mathbb{R}).$$

Domain of $f+g$. The sum $f+g$ is defined where both f and g are defined. Since

$$\mathrm{dom}(f) = \{x \in \mathbb{R} : x \le 4\}, \qquad \mathrm{dom}(g) = \mathbb{R},$$

we have

$$\mathrm{dom}(f+g) = \mathrm{dom}(f) \cap \mathrm{dom}(g) = \{x : x \le 4\}.$$

Domain of fg. Similarly,

$$fg(x) = x^2\sqrt{4-x}$$

is defined exactly when f is defined, so

$$\mathrm{dom}(fg) = \{x : x \le 4\}.$$

Domain of $f \circ g$. We compute

$$(f \circ g)(x) = f(g(x)) = f(x^2) = \sqrt{4-x^2}\,.$$

This is defined when

$$4 - x^2 \ge 0 \quad \Longleftrightarrow \quad -2 \le x \le 2.$$

Thus,

$$\mathrm{dom}(f \circ g) = \{x \in \mathbb{R} : -2 \le x \le 2\}.$$

Domain of $g \circ f$. We compute

$$(g \circ f)(x) = g(f(x)) = (\sqrt{4-x})^2 = 4 - x.$$

This requires only that $f(x)$ be defined, so

$$\mathrm{dom}(g \circ f) = \{x : x \le 4\}.$$

(b) Evaluating the compositions:

$$\begin{aligned} f \circ g(0) &= \sqrt{4 - 0^2} = 2, & g \circ f(0) &= (\sqrt{4 - 0})^2 = 4, \\ f \circ g(1) &= \sqrt{4 - 1^2} = \sqrt{3}, & g \circ f(1) &= (\sqrt{4 - 1})^2 = 3, \\ f \circ g(2) &= \sqrt{4 - 4} = 0, & g \circ f(2) &= (\sqrt{4 - 2})^2 = 2. \end{aligned}$$

(c) The two composite functions are

$$(f \circ g)(x) = \sqrt{4 - x^2}, \qquad (g \circ f)(x) = 4 - x.$$

These expressions are not equal as functions, so

$$f \circ g \neq g \circ f.$$

(d) The value $f \circ g(3)$ is not meaningful because $3 \notin \operatorname{dom}(f \circ g)$ (since $|3| > 2$). However, $g \circ f(3)$ *is* meaningful because $3 \leq 4$ lies in $\operatorname{dom}(g \circ f)$:

$$g \circ f(3) = \left(\sqrt{4 - 3}\right)^2 = 1.$$

$\square$

Problem 2 (ε–δ *Continuity of Real Valued Functions*)

Prove, using the ε-δ definition of continuity, Theorem 17.4 (i). Let f and g be real-valued functions that are continuous at x_0 in $\mathbb{R}$. Then $f+g$ is continuous at x_0.

Discussion.
To prove that $f+g$ is continuous at x_0 using the ε–δ definition, we begin by recalling the definition of continuity. A function h is continuous at x_0 if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|x - x_0| < \delta \quad \Longrightarrow \quad |h(x) - h(x_0)| < \varepsilon.$$

Since both f and g are continuous at x_0, we are guaranteed the existence of numbers δ_1 and δ_2 corresponding to f and g, respectively. The key intuition is that

$$|(f+g)(x) - (f+g)(x_0)| = |f(x) - f(x_0) + g(x) - g(x_0)|$$

can be controlled using the triangle inequality. By splitting the tolerance ε into two equal parts, $\varepsilon/2$ and $\varepsilon/2$, we may force each term

$$|f(x) - f(x_0)| \qquad \text{and} \qquad |g(x) - g(x_0)|$$

to be small simultaneously. To ensure this, we take

$$\delta = \min\{\delta_1, \delta_2\},$$

which guarantees that both inequalities hold whenever $|x - x_0| < \delta$. This approach is standard in proofs where the triangle inequality is used to break a sum into two individually controlled pieces. Thus, the continuity of $f+g$ at x_0 follows from the continuity of f and g at x_0.

Proof.
Fix $\varepsilon > 0$. Since f is continuous at x_0, there exists $\delta_1 > 0$ such that

$$|x - x_0| < \delta_1 \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon/2.$$

Since g is continuous at x_0, there exists $\delta_2 > 0$ such that

$$|x - x_0| < \delta_2 \quad \Longrightarrow \quad |g(x) - g(x_0)| < \varepsilon/2.$$

Let

$$\delta = \min\{\delta_1, \delta_2\}.$$

Now suppose $|x - x_0| < \delta$. Then $|x - x_0| < \delta_1$ and $|x - x_0| < \delta_2$, so the above inequalities apply. Using the triangle inequality,

$$\begin{aligned} |(f+g)(x) - (f+g)(x_0)| &= |f(x) - f(x_0) + g(x) - g(x_0)| \\ &\le |f(x) - f(x_0)| + |g(x) - g(x_0)| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, this proves that $f+g$ is continuous at x_0. □

Problem 3 (*Limit Theorems for Sequences Applied to Functions*)

Prove, using the limit theoerems for sequences, Thoerem 17.4 (ii). Let f and g be real-valued functions that are continuous at x_0 in $\mathbb{R}$. Then fg is continuous at x_0.

Discussion.
The goal is to prove part **(ii)** of Theorem 17.4 using *only* the limit theorems for sequences.

From earlier in the chapter we know the *sequential characterization of continuity*: a real-valued function h is continuous at x_0 iff for every sequence (x_n) in $\mathbb{R}$ with $x_n \to x_0$, we have

$$h(x_n) \to h(x_0).$$

In words: to test continuity at x_0, it is enough to see what h does to sequences that converge to x_0.

We also have the standard limit theorem for products of sequences: if (a_n) and (b_n) are sequences with $a_n \to A$ and $b_n \to B$, then

$$a_n b_n \to AB.$$

Putting these two facts together suggests the strategy:

- Start with an arbitrary sequence (x_n) with $x_n \to x_0$.
- Because f and g are continuous at x_0, the sequential definition tells us that

$$f(x_n) \to f(x_0) \quad \text{and} \quad g(x_n) \to g(x_0).$$

- Now we can view $\big(f(x_n)\big)$ and $\big(g(x_n)\big)$ as two sequences and apply the product limit theorem to obtain

$$f(x_n)g(x_n) \to f(x_0)g(x_0).$$

- Finally, observe that $f(x_n)g(x_n) = (fg)(x_n)$ and $f(x_0)g(x_0) = (fg)(x_0)$, so we have shown

$$(fg)(x_n) \to (fg)(x_0)$$

  for every sequence (x_n) with $x_n \to x_0$. By the sequential characterization, this is exactly the statement that fg is continuous at x_0.

So the whole proof is really about "passing to sequences," using continuity of f and g one at a time, then recombining them with the product theorem.

17.4 Theorem

Let f and g be real-valued functions that are continuous at x_0 in $\mathbb{R}$. Then

(i) $f + g$ is continuous at x_0;

(ii) fg is continuous at x_0;

(iii) f/g is continuous at x_0 if $g(x_0) \neq 0$.

Proof. Let f and g be real-valued functions that are continuous at x_0 in $\mathbb{R}$. We want to show that fg is continuous at x_0.

By the sequential characterization of continuity, it suffices to prove that for every sequence (x_n) in $\mathbb{R}$ with $x_n \to x_0$, we have

$$(fg)(x_n) \to (fg)(x_0).$$

So let (x_n) be an arbitrary sequence with $x_n \to x_0$. Since f is continuous at x_0, the sequence $(f(x_n))$ satisfies

$$f(x_n) \to f(x_0).$$

Similarly, since g is continuous at x_0, the sequence $(g(x_n))$ satisfies

$$g(x_n) \to g(x_0).$$

Now apply the product limit theorem for sequences: because $f(x_n) \to f(x_0)$ and $g(x_n) \to g(x_0)$, we obtain

$$f(x_n)g(x_n) \to f(x_0)g(x_0).$$

But $f(x_n)g(x_n) = (fg)(x_n)$ and $f(x_0)g(x_0) = (fg)(x_0)$, so we have shown

$$(fg)(x_n) \to (fg)(x_0)$$

for every sequence (x_n) with $x_n \to x_0$.

By the sequential characterization of continuity, this means that fg is continuous at x_0. $\square$

Problem 4 (*Ross Chapter 3 17.8*)

Note: Example 5 on page 130 may be helpful. Let f and g be real-valued functions.

(a) Show $\min(f,g) = \frac{1}{2}(f+g) - \frac{1}{2}|f-g|$.

(b) Show $\min(f,g) = -\max(-f,-g)$.

(c) Use (a) or (b) to prove that if f and g are continuous at x_0 in $\mathbb{R}$, then $\min(f,g)$ is continuous at x_0.

Discussion.
In Example 5, Ross shows

$$\max(f,g) = \tfrac{1}{2}(f+g) + \tfrac{1}{2}|f-g|$$

and then proves continuity of $\max(f,g)$ by breaking it into pieces built from f and g using the sum, product, and absolute–value theorems.

Here we are doing the same game, but with the *minimum* instead of the maximum. Parts (a) and (b) are purely algebraic identities: first we show that for real numbers a, b,

$$\min\{a,b\} = \tfrac{1}{2}(a+b) - \tfrac{1}{2}|a-b| \qquad \text{and} \qquad \min\{a,b\} = -\max\{-a,-b\}.$$

Those formulas are then applied pointwise with $a = f(x)$ and $b = g(x)$.

For part (c) there are two natural strategies:

- Use part (a) directly and imitate Example 5, writing $\min(f,g)$ as a combination of $f+g$ and $|f-g|$, both of which are continuous when f and g are.
- Use part (b) together with Example 5: if f and g are continuous, then so are $-f$ and $-g$, so Example 5 tells us that $\max(-f,-g)$ is continuous; then $\min(f,g) = -\max(-f,-g)$ is just -1 times a continuous function.

I'll use both approaches in the algebraic parts, and then for continuity in (c) I will lean on part (b) plus Example 5, since that's exactly what Ross had in mind.

Proof.
(a) We first prove the identity for real numbers and then apply it to functions. Let $a, b \in \mathbb{R}$ and define

$$m := \frac{1}{2}(a+b) - \frac{1}{2}|a-b|.$$

We show that $m = \min\{a,b\}$ by considering the two cases $a \geq b$ and $a < b$.

Case 1: $a \geq b$. Then $|a-b| = a-b$, so

$$m \;=\; \frac{1}{2}(a+b) - \frac{1}{2}(a-b) \;=\; \frac{1}{2}(a+b-a+b) \;=\; \frac{1}{2}(2b) \;=\; b.$$

Since $a \geq b$, the minimum of $\{a, b\}$ is b, so in this case $m = \min\{a, b\}$.

Case 2: $a < b$. Then $|a - b| = b - a$, so

$$m \;=\; \frac{1}{2}(a+b) - \frac{1}{2}(b-a) \;=\; \frac{1}{2}(a+b-b+a) \;=\; \frac{1}{2}(2a) \;=\; a.$$

Now $a < b$, so the minimum of $\{a, b\}$ is a, and again $m = \min\{a, b\}$.

Since these are the only two possibilities, we have proved

$$\min\{a, b\} = \frac{1}{2}(a+b) - \frac{1}{2}|a-b| \quad \text{for all } a, b \in \mathbb{R}.$$

Replacing a by $f(x)$ and b by $g(x)$ gives

$$\min(f, g)(x) = \frac{1}{2}\big(f(x) + g(x)\big) - \frac{1}{2}\big|f(x) - g(x)\big|$$

for all x, which is exactly

$$\min(f, g) = \tfrac{1}{2}(f+g) - \tfrac{1}{2}|f-g|.$$

(b) Again start with real numbers $a, b \in \mathbb{R}$. We claim

$$\min\{a, b\} = -\max\{-a, -b\}.$$

Consider the same two cases.

Case 1: $a \leq b$. Then $-a \geq -b$, so

$$\max\{-a, -b\} = -a \quad \Rightarrow \quad -\max\{-a, -b\} = -(-a) = a.$$

Since $a \leq b$, the minimum of $\{a, b\}$ is a, so $\min\{a, b\} = -\max\{-a, -b\}$.

Case 2: $a > b$. Then $-a < -b$, so

$$\max\{-a, -b\} = -b \quad \Rightarrow \quad -\max\{-a, -b\} = -(-b) = b.$$

Here $\min\{a, b\} = b$, so once again $\min\{a, b\} = -\max\{-a, -b\}$.

Thus the identity holds for all $a, b \in \mathbb{R}$. Substituting $a = f(x)$ and $b = g(x)$ yields

$$\min(f, g)(x) = -\max(-f, -g)(x),$$

that is,

$$\min(f, g) = -\max(-f, -g).$$

(c) Assume now that f and g are continuous at $x_0 \in \mathbb{R}$. We prove that $\min(f, g)$ is continuous at x_0.

By continuity and Theorem 17.3, the functions $-f$ and $-g$ are continuous at x_0 (they are just constant multiples of f and g). Then, by Example 5, the function $\max(-f, -g)$ is continuous at x_0.

From part (b) we have the identity

$$\min(f, g) = -\max(-f, -g).$$

The right-hand side is -1 times a continuous function, hence continuous at x_0 by Theorem 17.3 again.

Therefore $\min(f, g)$ is continuous at x_0. $\square$

Problem 5 (*Ross Chapter 3 17.12*)

(a) Let f be a continuous real-valued function with domain (a, b). Show that if $f(r) = 0$ for each rational number r in (a, b), then $f(x) = 0$ for all $x \in (a, b)$.

(b) Let f and g be continuous real-valued functions on (a, b) such that $f(r) = g(r)$ for each rational number r in (a, b). Prove $f(x) = g(x)$ for all $x \in (a, b)$. *Hint:* Use part (a).

Discussion.
Part (a) is really about how much mileage we get out of continuity together with the density of the rational numbers in (a, b). Intuitively, if f is continuous and vanishes at every rational point, then it has no room to "jump up" at an irrational point: any irrational x can be approached by a sequence of rationals $r_n \to x$, and continuity forces

$$f(x) = \lim_{n\to\infty} f(r_n) = \lim_{n\to\infty} 0 = 0.$$

So the strategy is:

- Fix an arbitrary $x \in (a, b)$.
- If x is rational, we already know $f(x) = 0$ by hypothesis.
- If x is irrational, use the density of $\mathbb{Q}$ to choose a sequence of rational numbers (r_n) with $r_n \in (a, b)$ and $r_n \to x$. Since f is continuous, $f(r_n) \to f(x)$; but each $f(r_n) = 0$, so the limit must be 0, forcing $f(x) = 0$.

Part (b) is then a simple and elegant reuse of (a). If f and g are continuous and agree on all rationals, we consider the difference

$$h := f - g.$$

This h is continuous on (a, b) (by the algebra of continuous functions), and on every rational r we have $h(r) = f(r) - g(r) = 0$. By part (a), $h(x) = 0$ for all $x \in (a, b)$, which is just another way of saying $f(x) = g(x)$ everywhere on (a, b). So the hard work is done in (a); (b) is an application of that general principle to the difference of two functions.

Proof.
(a) Let f be a continuous real-valued function with domain (a, b), and assume $f(r) = 0$ for every rational $r \in (a, b)$. We must show that $f(x) = 0$ for all $x \in (a, b)$.

Let $x \in (a, b)$ be arbitrary.

Case 1: x is rational. Then, by hypothesis, $f(x) = 0$.

Case 2: x is irrational. Because the rational numbers are dense in $\mathbb{R}$, there exists a sequence (r_n) of rational numbers in (a, b) such that $r_n \to x$ as $n \to \infty$. By assumption, each r_n is rational, so

$$f(r_n) = 0 \quad \text{for all } n.$$

Hence

$$\lim_{n\to\infty} f(r_n) = 0.$$

On the other hand, f is continuous at x, and (r_n) is a sequence with $r_n \to x$, so by the sequential characterization of continuity,

$$\lim_{n\to\infty} f(r_n) = f(x).$$

Combining these equalities gives $f(x) = 0$.

Since $x \in (a, b)$ was arbitrary, we have shown that $f(x) = 0$ for all $x \in (a, b)$, as required.

(b) Let f and g be continuous real-valued functions on (a, b) such that $f(r) = g(r)$ for every rational $r \in (a, b)$. Define

$$h(x) := f(x) - g(x) \quad \text{for } x \in (a, b).$$

By Theorem 17.4(i), h is continuous on (a, b), since it is the sum of f and $-g$.

For each rational $r \in (a, b)$ we have

$$h(r) = f(r) - g(r) = 0.$$

Thus h is a continuous function on (a, b) that vanishes at every rational in (a, b). By part (a) applied to h, it follows that

$$h(x) = 0 \quad \text{for all } x \in (a, b).$$

Therefore $f(x) - g(x) = 0$ for all $x \in (a, b)$, i.e.,

$$f(x) = g(x) \quad \text{for all } x \in (a, b).$$

$\square$

Problem 6 (*Ross Chapter 17.13 (b)*)

Let $h(x) = x$ for rational numbers x and $h(x) = 0$ for irrational numbers. Show h is continuous at $x = 0$ and at no other point.

Discussion.
The definition of h tells us that it "remembers" the input x if x is rational, and drops it to 0 if x is irrational. Since rationals and irrationals are both dense in $\mathbb{R}$, every neighborhood of any point contains lots of points of each type. Intuitively, this suggests that h should jump around wildly near most points, because in any small interval we see values of h equal to x (at rational x) and equal to 0 (at irrational x). The one exception is at $x = 0$: there $h(0) = 0$, and near 0 the actual values $h(x)$ are either x or 0, both of which are small in absolute value when $|x|$ is small. That gives us hope that we can make $|h(x) - h(0)|$ as small as we like by forcing $|x|$ to be small.

To turn this into a proof we will split the problem into two parts:

- Show h is continuous at 0 using the ε–δ definition. The key inequality will be $|h(x) - h(0)| \leq |x|$.
- Show h is *not* continuous at any $x_0 \neq 0$ using the sequential characterization of continuity: we will construct, for each $x_0 \neq 0$, a sequence (x_n) converging to x_0 such that $\lim h(x_n) \neq h(x_0)$. Because both $\mathbb{Q}$ and its complement are dense, we can always choose a sequence of rationals or irrationals approaching x_0, whichever is more convenient.

Proof.
We first show that h is continuous at $x = 0$. Note that $h(0) = 0$ because 0 is rational. Let $\varepsilon > 0$ be given. Set $\delta = \varepsilon$. If $|x - 0| < \delta$, then $|x| < \delta = \varepsilon$. By the definition of h, either $h(x) = x$ (if x is rational) or $h(x) = 0$ (if x is irrational). In either case,

$$|h(x) - h(0)| = |h(x)| \leq |x| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this shows that h is continuous at $x = 0$.

Now let $x_0 \neq 0$. We will show that h is not continuous at x_0 by producing a sequence (x_n) with $x_n \to x_0$ but $\lim h(x_n) \neq h(x_0)$. We consider two cases.

Case 1: $x_0 \in \mathbb{Q}$. Define

$$x_n := x_0 + \frac{\sqrt{2}}{n}, \qquad n \in \mathbb{N}.$$

Then $x_n \to x_0$ as $n \to \infty$. We claim that each x_n is irrational. Indeed, if some x_n were rational, then

$$\frac{\sqrt{2}}{n} = x_n - x_0$$

would be the difference of two rationals, hence rational, which would imply $\sqrt{2}$ is rational—a contradiction. Thus every x_n is irrational, so $h(x_n) = 0$ for all n, and therefore

$$\lim_{n\to\infty} h(x_n) = 0.$$

On the other hand, x_0 is rational and $x_0 \neq 0$, so

$$h(x_0) = x_0 \neq 0 = \lim_{n\to\infty} h(x_n).$$

By the sequential characterization of continuity, h is not continuous at this rational x_0.

Case 2: $x_0 \notin \mathbb{Q}$. Because the rationals are dense in $\mathbb{R}$, there exists a sequence (r_n) of rational numbers such that $r_n \to x_0$ as $n \to \infty$. For each n we then have

$$h(r_n) = r_n,$$

so

$$\lim_{n\to\infty} h(r_n) = \lim_{n\to\infty} r_n = x_0.$$

But x_0 is irrational, so by definition $h(x_0) = 0$. Hence

$$\lim_{n\to\infty} h(r_n) = x_0 \neq 0 = h(x_0).$$

Again, by the sequential characterization of continuity, h is not continuous at this irrational x_0.

In both cases we have shown that h fails to be continuous at every $x_0 \neq 0$, while it is continuous at $x_0 = 0$. Therefore h is continuous exactly at $x = 0$ and at no other point. $\square$